Audio-Augmented Prompting for Humor Explanation in Open-Source Language Models

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***Abstract*—Humor understanding remains difficult for large language models despite strong performance on many text-based language tasks. Humor often depends on spoken delivery as well as wording, including prosody and intonation. This paper investigates a simple multimodal prompting approach for humor explanation in which a model receives both the joke text and a spoken rendition produced by an off-the-shelf text-to-speech system. We compare a text-plus-audio condition against a text-only baseline across multiple humor datasets and evaluate the quality of generated explanations using an independent judging model. The results show that adding the spoken signal can improve explanation quality for pronunciation-dependent humor, while providing mixed benefits for broader humor that does not rely on phonetic ambiguity. The proposed setup is lightweight and can be applied without changing model architecture or performing task-specific training.**

***Index Terms*—humor understanding, multimodal prompting, text-to-speech, prosody, explanation generation**

I. Introduction

Recent advances in large language models have enabled strong performance across a wide range of natural language understanding tasks [1]. However, humor understanding remains challenging, particularly in settings where linguistic ambiguity and phonetic cues are central. Similar observations have been made in related multimodal affective language tasks such as sarcasm recognition, where integrating textual and acoustic cues improves performance [2]. More broadly, linguistic ambiguity itself is a well-known source of humor in everyday language, including lexical and syntactic ambiguity in short texts such as newspaper headlines [3]. In pun-based humor, pronunciation and prosody can provide information that is not always recoverable from written text alone.

II. Related Work

Computational humor research has long emphasized ambiguity as a central mechanism underlying jokes and puns. Early linguistic theories characterize humor as arising from the coexistence of multiple semantic interpretations within a single utterance [4]. Building on this perspective, formal semantic and pragmatic analyses of puns highlight the role of phonological similarity and lexical ambiguity in enabling dual readings [5]. In computational settings, humor detection and interpretation have been explored using supervised and neural approaches, including benchmarks such as SemEval-2017 Task 7, which operationalizes pun detection and sense interpretation in English text [8].

More recently, large language models have demonstrated strong capabilities in general natural language understanding, yet their performance on humor remains inconsistent, particularly when jokes rely on subtle ambiguity or world knowledge [6]. Parallel developments in multimodal language understanding suggest that integrating acoustic information can improve performance in tasks such as emotion recognition and spoken language understanding, where prosody and delivery provide complementary cues to text [7]. These findings motivate examining whether similar multimodal benefits extend to pronunciation-dependent humor, especially under reproducible, locally deployable model settings.

III. Methods

A controlled evaluation is conducted to determine whether access to audio information improves humor understanding and explanation in locally deployed open-source language models. Given the limited capacity of such models relative to large proprietary systems, a simple multimodal prompting framework is adopted. Few-shot exemplars and explicit chain-of-thought prompting are not used. Instead, all experiments are performed in a zero-shot setting with controlled output formatting.

As illustrated in Fig. 1, the pipeline is organized into four stages: (1) explanation generation from text-only input, (2) explanation generation from combined text and audio input, (3) post-processing and normalization of model outputs into a strict JSON representation, and (4) comparative evaluation using a separate judging model.

A. Audio generation

Spoken realizations of all jokes are generated to expose pronunciation-related cues relevant to humor understanding. An off-the-shelf text-to-speech system is employed. Specifically, Piper TTS is used as a lightweight open-source engine suitable for local deployment, and audio is synthesized directly from the original joke text. No additional supervision is provided, including emphasis, timing, or prosody annotations. The synthesized audio is used only as an auxiliary input intended to reveal pronunciation-based ambiguity.

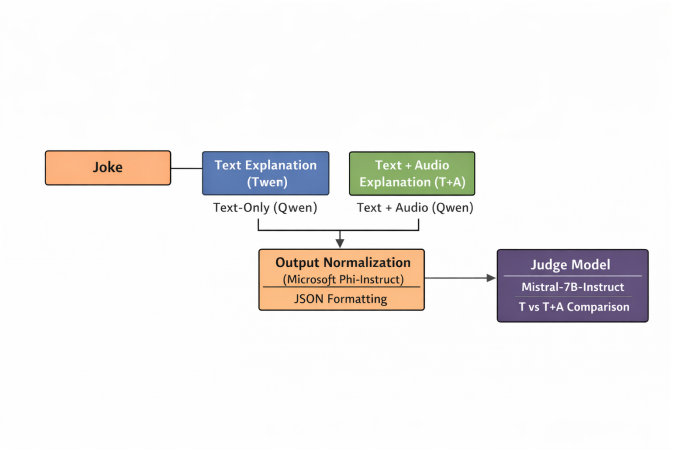

Fig. 1. Pipeline overview: text-only and text-plus-audio explanation generation, JSON normalization, and judge evaluation.

B. Prompting

All explanation prompts are formulated in a zero-shot configuration. Each prompt includes a concise definition of puns and non-puns, together with instructions to determine whether the input constitutes a pun and to describe the linguistic mechanism when applicable. Few-shot examples and explicit chain-of-thought prompting are omitted because preliminary trials indicate that smaller local models do not produce stable and consistent outputs under these settings.

Two explanation conditions are evaluated:

- **Text-Only:** the model receives only the written joke.
- **Text-Plus-Audio:** the model receives the written joke and its synthesized spoken form within a single prompt.

To ensure compatibility with text-only ground truth annotations, the prompt instructs the model to avoid referencing the presence or absence of audio in the generated explanation.

C. Output normalization and evaluation

All generated explanations are converted to a strict JavaScript Object Notation (JSON) format to enforce consistency and enable automated comparison. Outputs that do not conform to the required format are marked as invalid.

To assess explanation quality, a separate judging model is employed. For pairwise comparisons, the judge receives the original joke text and the two explanations produced under the text-only and text-plus-audio conditions and determines which explanation more accurately identifies whether the input constitutes a pun and more effectively explains the linguistic mechanism. The judging instructions discourage positional bias and verbosity bias and emphasize correctness, clarity, and accurate identification of wordplay. A tie decision is permitted when both explanations are comparable. No audio input is provided to the judging model.

IV. Datasets and Models

The proposed approach is evaluated on three humor datasets that differ in dataset size, annotation quality, and the availability of human-written explanations.

1) SemEval 2017 Task 7: The dataset [8] comprises 810 homographic puns, 647 heterographic puns, and 1,077 non-puns, with pun instances explicitly partitioned by type. This structure makes the dataset well-suited for evaluating explanation quality and for analyzing the linguistic mechanisms underlying pun-based humor.

2) Context-Situated Puns: The Context-Situated Puns dataset contains 821 homographic puns and 1,739 heterographic puns with human annotations specifying the pun word and its intended interpretations. Each instance consists of a single sentence containing a pun, supporting evaluation of whether the model can infer and explain the underlying linguistic ambiguity from sentence-level context [9].

3) ExplainTheJoke: To evaluate explanation quality beyond pun-based humor, the ExplainTheJoke dataset is used in its entirety. The dataset contains jokes spanning multiple humor categories and does not include structured annotations. For each joke, the judge evaluates the text-only explanation and the text-plus-audio explanation *independently* on a five-point scale based on correctness, clarity, and accurate identification of the humor mechanism, while penalizing hallucinated or irrelevant reasoning. No audio input is provided to the judging model. Pairwise preferences are then derived by comparing the two scores; equal scores are counted as ties.

A. Models

All experiments are conducted using freely available locally deployable language models.

Explanation generation: Both text-only and text-plus-audio explanations are generated using Qwen2-Audio, which supports audio-conditioned generation while remaining deployable in a local environment. The same model and prompt structure are used for both conditions to ensure a controlled comparison, with the only difference being the presence or absence of audio input.

Output normalization: Generated explanations are normalized into a strict JSON format using Microsoft Phi Instruct. Outputs that fail to conform to the required format are marked as invalid and excluded from further analysis.

Evaluation judge model: Explanation quality is assessed using Mistral 7B Instruct v0.3, employed as an independent judging model. Using a separate model for evaluation reduces the risk of a model judging its own outputs.

V. Results

A. SemEval 2017 Task 7

Evaluation on SemEval is performed using the comparison-based judging framework. A balanced subset of 250 homographic and 250 heterographic puns is sampled. For each joke, two explanations are generated, one using text-only prompting and one using text-plus-audio prompting. The distribution of preferences, ties, and invalid outputs is reported in Table I.

B. Context-Situated Puns

For Context-Situated Puns, evaluation is conducted using the same judging framework, but explanations are scored

TABLE I
SEMEVAL RESULTS UNDER PAIRWISE JUDGING (VALID OUTPUTS ONLY).

Split (Valid N)	Text wins	Audio wins	Tie
Heterographic (188)	41.5%	58.0%	0.5%
Homographic (194)	47.4%	49.5%	3.1%

TABLE II
RESULTS ON CONTEXT-SITUATED PUNS. PREFERENCES ARE DERIVED BY COMPARING THE JUDGE-ASSIGNED 1–5 SCORES FOR TEXT-ONLY VS. TEXT-PLUS-AUDIO EXPLANATIONS; EQUAL SCORES ARE COUNTED AS TIES.

Outcome (Valid N=299)	Count	Percent
Tie (scores equal)	232	77.6%
Text > Audio	53	17.7%
Audio > Text	14	4.7%
Score statistics (1–5)	**Mean**	**Median**
Text-only	3.696	3.0
Text + Audio	3.428	3.0

independently rather than compared directly. A subset of 300 examples is sampled, covering both homographic and heterographic puns. For each example, two explanations are generated under the text-only and text-plus-audio conditions. The judging model assigns each explanation a score on a five-point scale (1–5) based on correctness, clarity, and accurate identification of the linguistic mechanism. Pairwise preferences are then derived by comparing the two scores, with equal scores counted as ties.

C. ExplainTheJoke

For ExplainTheJoke, the judge scores each explanation independently on a five-point scale. A pairwise preference is then derived by selecting the explanation with the higher score; equal scores are counted as ties. Table III reports the resulting preference distribution.

TABLE III
EXPLAINTHEJOKE RESULTS. THE JUDGE SCORES EACH EXPLANATION INDEPENDENTLY (1–5). PREFERENCES ARE DERIVED BY COMPARING THE TWO SCORES; EQUAL SCORES ARE COUNTED AS TIES.

Derived preference	Count	Percent
Text-only (score higher)	63	16.7%
Text + Audio (score higher)	39	10.3%
Tie (scores equal)	275	72.9%

VI. DISCUSSION AND CONCLUSION

Across datasets, the effect of adding audio is mixed and appears to depend on humor type and the stability of local generation. Pronunciation-dependent cases, such as heterographic puns, are more likely to benefit from audio, because the spoken form can expose phonetic ambiguity that may be underspecified in text. In contrast, humor that does not rely on phonetic cues can show limited or inconsistent benefits, and synthesized speech may introduce noise when prosody does not add discriminative information.

Taken together, these findings suggest that multimodal prompting with locally deployable models can help in pronunciation-dependent settings, but is not universally beneficial. Future work should explore higher quality speech synthesis, controlled prosody, and prompt designs that better exploit audio information while maintaining stable output formatting.

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